

A Mechanistic Refusal Direction in a Small Instruct Model

The refusal decision is linearly legible in last-prompt-token activation *norms* (AUROC 0.98, leave-one-out 0.825, permutation $p=0.005$) where the first-token logit margin is not

hexa-codex maintainers
github.com/dancinlab/hexa-codex

May 25, 2026

Abstract

Per the hexa-codex `cx_paper_gate` workflow rule, a paper ships only when every section claim reaches a 100% formal-blue or numerical-green verdict. We report the SAFETY-domain finding measured on a self-hosted full-precision `Qwen2.5-1.5B-Instruct` [15] (NVIDIA RTX 5070, HuggingFace `transformers` forward-pass hooks [14]): on a variance-controlled matched-pair adversarial-vs-benign set (20 adversarial + 20 benign-matched prompts, 20 refused / 20 answered by a fixed marker scan), the per-layer residual/attention/MLP activation-*norm* vector at the last prompt token (84 features = 28 layers $\times$ 3 sublayer kinds) *linearly separates* refused from answered. A difference-of-means refusal direction w yields a full-vector projection AUROC of 0.98; a leave-one-out (held-out) linear classifier reaches 0.825 (33/40) against a majority baseline of 0.50; and a 200-shuffle permutation test gives $p=0.005$ (0/200 label shuffles reach the observed accuracy). The same surface read through the first-token *logit* margin is a confirmed *negative* (the refused-vs-answered gap sits $5.9\times$ below the bimodality bar): the refusal decision is mechanistically legible in activation norms but *not* in the next-token logprob margin. The direction tracks the refusal *decision*, not the adversarial-vs-benign topic — the single adversarial-but-*answered* prompt projects to the answered side. **Scope (not claimed):** $n=40$ is small and the in-sample 84-dim AUROC can overfit; the load-bearing numbers are therefore the held-out leave-one-out accuracy and the permutation p , both reproduced by an independent recompute. We do *not* claim a single-direction ablation, a full sparse-autoencoder feature decomposition, or transfer to other models (those are deferred). The §Formula recompute (`verify/numerics_safety_refusal_direction.hexa`) passes 5/5 checks against the committed numbers-only matrix; verdict `.verdicts/sandbox/m4_safety_refusal.di`

1 Background: mechanistic interpretability and the linear hypothesis

The behaviour of a transformer language model is carried by its *residual stream* — the running sum, one vector per token position, to which every attention and MLP sublayer writes its output [6]. *Mechanistic interpretability* is the program of reading that internal state as a computation rather than treating the model as a black box. Two empirical regularities make the program tractable. The first is the *linear representation hypothesis*: many human-meaningful attributes are encoded as approximately linear directions in activation space, so that the presence of a concept corresponds to a displacement along one vector and can be read out by a dot product [11, 12]. The second is that the linear separability of such attributes *increases with depth* — a fact first quantified by Alain and Bengio [1], who trained a linear “probe” classifier on each layer’s activations and showed accuracy rising monotonically through the stack.

A linear probe is exactly the instrument we use. Rather than asking a model to introspect, one fixes the weights, runs a forward pass on a labelled set, extracts the activations at a chosen

site, and fits a simple linear classifier; its accuracy lower-bounds how legibly the attribute is encoded at that site [1, 3]. The probe is *diagnostic*: a direction that separates two behaviours is evidence the model represents the distinction linearly, though — absent an intervention — it does not by itself prove the direction is *causally* used (we return to this in §8 and §9). The present paper applies this classical probe methodology to the *refusal* behaviour of a small instruct model, with one deliberate restriction: our feature is not the dense hidden vector but its per-sublayer L_2 norm, a 84-dimensional summary that keeps the adversarial prompt content out of the committed artifact (§4).

2 Related work

The refusal direction. The directly adjacent result is Arditi et al. [2], who show across 13 open chat models (up to 72B) that refusal is mediated by a *single* one-dimensional subspace of the residual stream: erasing that direction suppresses refusal of harmful instructions, and adding it elicits refusal even on harmless ones — an *interventional* (causal) demonstration. Our finding is the small-model, norm-space, *correlational* analogue. We do not re-establish their causal claim; we establish, on a different and smaller model (Qwen2.5-1.5B-Instruct, not in their suite) and a deliberately reduced feature (per-sublayer activation *norms*, not the dense residual vector), that a linear refusal *direction* is recoverable and held-out predictive, and we localise it to three mechanistically distinct activation-norm motifs (§5). The causal single-direction ablation of Arditi et al. [2] on *our* surface is precisely the deferred M5.SAFETY residual (2) (§8).

Activation steering. The complementary control technique is *activation engineering*: computing a steering vector by contrasting activations on paired prompts and adding it during the forward pass to shift a high-level output property. Turner et al. [13] formalise this as *Activation Addition* (ActAdd), deriving a steering vector from a single contrastive pair and steering sentiment/topic at inference with no fine-tuning. The mean-difference direction w of §3 is the same contrastive object computed at scale (20 vs 20, in z -scored norm space) and used here as a *read-out* rather than a *write-in*; an ActAdd-style write-in test is one form the deferred causal residual could take.

Dictionary learning and sparse autoencoders. A richer account of a direction asks whether it decomposes into named, monosemantic *features*. Sparse autoencoders (SAEs) attack the polysemanticity / superposition problem by reconstructing activations as a sparse combination of an over-complete feature dictionary: Cunningham et al. [5] show such features are more interpretable and more causally precise than raw neuron or PCA directions, and Bricken et al. [4] scale dictionary learning to thousands of monosemantic features in a one-layer model. Our norm-space direction is explicitly *not* an SAE feature atlas — it establishes that a refusal direction *exists* and is held-out predictive in a compact norm summary; the full SAE decomposition of the three motifs is the other half of the deferred M5.SAFETY residual.

Representation reading. More broadly, representation-engineering [16] reads and controls model behaviour through population-level activation directions; the difference-of-means refusal direction is one concrete instance. Our methodological contribution within this line is not a new technique but an *honesty discipline*: the entire finding is recomputed from a committed numbers-only matrix by an independent verifier (§3), with the load-bearing claim placed on a held-out leave-one-out accuracy and a permutation p rather than the in-sample AUROC, and with the adversarial prompt text never leaving a private set (`cx_hf_safety_private`).

3 Formula

SUPPORTED-NUMERICAL — the classifier is a *difference-of-means linear direction* in a standardized activation-norm feature space, the norm-space analogue of the linear refusal direction reported for activation *vectors* [2] and of representation reading more broadly [16]. Let $a_i \in \mathbb{R}^{84}$ be the last-prompt-token activation-norm vector for prompt i (feature j is the L_2 norm of one (layer, kind) hidden state, $\text{kind} \in \{\text{residual}, \text{attn}, \text{mlp}\}$), and let $y_i \in \{0, 1\}$ be its refusal label (1 = refused). Each feature is z -scored over the $n=40$ rows,

$$z_{ij} = \frac{a_{ij} - \mu_j}{\sigma_j}, \quad \mu_j = \frac{1}{n} \sum_i a_{ij}, \quad \sigma_j = \sqrt{\frac{1}{n} \sum_i (a_{ij} - \mu_j)^2}, \quad (1)$$

(population σ , with the degenerate $\sigma_j=0$ replaced by 1). The refusal direction and per-row projection score are

$$w_j = \text{mean}_{i: y_i=1} z_{ij} - \text{mean}_{i: y_i=0} z_{ij}, \quad s_i = \sum_{j=1}^{84} z_{ij} w_j, \quad (2)$$

and the refuse-vs-answer rule is $\hat{y}(a) = \mathbb{I}[w \cdot z(a) > \theta]$, with θ the midpoint of the two class-projection means. Three threshold-free / held-out statistics certify the direction [7, 9]: (A) the full-vector projection AUROC (rank statistic, refused = positive); (B) the leave-one-out held-out linear accuracy (refit w on the other $n-1$ rows, threshold at the train class-mean midpoint, predict the held-out row); (C) a 200-shuffle permutation p on the leave-one-out accuracy.

Why the difference of means is the right linear direction. The choice of $w = \text{mean}(z_{\text{refused}}) - \text{mean}(z_{\text{answered}})$ is not ad hoc: it is the *Bayes-optimal* linear discriminant under a standard generative assumption, and it degrades gracefully when that assumption is only approximate. Model each class’s standardized feature vector as Gaussian with a *shared* covariance Σ and class means μ_1 (refused), μ_0 (answered), $z \mid y \sim \mathcal{N}(\mu_y, \Sigma)$. The log-likelihood ratio is

$$\log \frac{p(z \mid y=1)}{p(z \mid y=0)} = \underbrace{(\mu_1 - \mu_0)^\top \Sigma^{-1}}_{w_{\text{LDA}}^\top} z + \text{const}, \quad (3)$$

the quadratic terms cancelling exactly *because* the covariance is shared. The optimal decision is therefore a *threshold on a single linear projection* $w_{\text{LDA}}^\top z$, with $w_{\text{LDA}} = \Sigma^{-1}(\mu_1 - \mu_0)$: Fisher’s linear discriminant [8, 10]. Our $w = \mu_1 - \mu_0$ is the *isotropic* ($\Sigma \propto I$) special case — the Bayes-optimal direction when the standardized features are uncorrelated and equal-variance, which the per-feature z -scoring of Eq. (1) makes a deliberate, transparent modelling choice. Two consequences matter here. (i) With $n=40$ rows and 84 features the empirical Σ is rank-deficient and its inverse is *not* estimable; using the diagonal-isotropic surrogate $w = \mu_1 - \mu_0$ *avoids fitting* $\binom{84}{2}$ covariance parameters from 40 samples — it is the high-bias, low-variance end of the discriminant family and exactly the right end at this sample size (this is also why we never claim the in-sample AUROC; §8). (ii) The direction is the population contrast used by activation-steering methods [13]: the same $\mu_1 - \mu_0$ object, read out here rather than added in.

Projection and threshold geometry. Projecting a row onto w gives the scalar score $s_i = w^\top z_i$; under Eq. (3)’s isotropic model the two classes’ scores are univariate Gaussians whose means are $\bar{s}_1 = w^\top \mu_1$ and $\bar{s}_0 = w^\top \mu_0$ with $\bar{s}_1 - \bar{s}_0 = \|\mu_1 - \mu_0\|^2 \geq 0$ — the refused class sits, by construction, at the higher projection. The Bayes threshold for equal priors and equal projected variances is the *midpoint* $\theta = (\bar{s}_1 + \bar{s}_0)/2$, which is exactly the rule used in code (and which the recompute reproduces as $\theta \approx 0$ by symmetry of the z -scored, balanced 20/20 design, with class-projection means ± 24.61). Geometrically, w is the *normal* of the separating hyperplane

and s_i is the signed distance (up to the constant $\|w\|$) of row i from it; the AUROC is the probability that a random refused row out-projects a random answered row, a threshold-free read of how far apart the two score Gaussians sit.

Recompute. The verifier `verify/numerics_safety_refusal_direction.hexa` reads the committed numbers-only matrix `.verdicts/sandbox/m2_safety_refusal_norms.tsv` (no model spawn, no GPU, no API) and recomputes w , the projections, and all three statistics, checking five conditions (all pass): (1) matrix shape 40×84 , balanced 20/20; (2) projection AUROC reproduces 0.98 (drift 0.0); (3) leave-one-out accuracy reproduces 0.825 (33/40, drift 0.0) and exceeds the majority 0.50; (4) permutation $p \leq 0.05$ (0/200 shuffles reach the observed accuracy, $p=0.00498$); (5) the topic-confound control — the lone adversarial-but-answered row projects to the answered side ($s=-25.81$, matching the original probe’s -25.81). Verdict `.verdicts/sandbox/m4_safety_refusal_direction_recompute.txt` (5/5).

Permutation-test honesty (explicitly disclosed). The original probe shuffled labels with Python’s Mersenne-Twister (seed 1234); the hexa verifier uses its own fixed-seed linear-congruential Fisher–Yates shuffle for the same 200-permutation leave-one-out test. The test *design* is identical, the pseudo-random stream differs. The observed accuracy (0.825) sits so far above the chance band that 0/200 shuffles reach it under either generator, so $p=(0+1)/(200+1) \approx 0.005$. The p -value’s authority is the permutation design and the large margin, not a specific RNG implementation.

4 Method

SUPPORTED-NUMERICAL — activations are captured by the hook logic of `lm_foundation/tool/activation_capture.py` (the HuggingFace `transformers` `register_forward_hook` backend), run on a self-hosted NVIDIA RTX 5070 (12 GB) host `ubu-1` serving *full-precision* (fp32) Qwen2.5-1.5B-Instruct [15] — the true weights, *not* a quantized GGUF — in a clean pinned virtual environment (`transformers==4.51.3`, `numpy<2`). For every prompt a forward hook on each of the 28 decoder layers records the L_2 norm of the residual-stream output, the self-attention output, and the MLP output *at the last prompt-token position* (the position whose hidden state forms the answer/refuse decision before generation), giving the ordered 84-feature vector. The refusal label is an exact reuse of the prior verdicts’ marker scan: a short 48-token greedy generation (`do_sample=False`, fp32) is labelled **refused** iff any of 24 fixed refusal markers (e.g. “i can’t”, “i’m sorry”, “i must decline”) appears in the first 200 characters — *no* LLM self-judge, per the SANDBOX honesty rule.

Test set (pre-registered, variance-controlled). The prompts are the matched-pair set of `bench/sandbox_stage4_refusal_bimodal_tighter.hexa` (HuggingFace private `dancinlab/hexa-codex-sandbox` adversarial sets default private per `cx_hf_safety_private`): 4 domains (social, physical-safety, self-harm, medical) $\times$ 5 pairs, each adversarial prompt paired with a benign prompt of the *same* domain, verb, approximate length, and prose-answer regime, so the only systematic adversarial–benign difference is the safety trigger. The observed label distribution is refused=20 / answered=20 (adv_refused=19/20, benign_refused=1/20) — clean specificity, with two honest label-flipped rows kept as the topic-confound control of §3. The committed analysis surface `.verdicts/sandbox/m2_safety_refusal_norms.tsv` contains the 40×84 activation norms and the binary labels *only*; the adversarial prompt strings are redacted per `cx_hf_safety_private`. A deterministic re-run of the forward pass reproduced the committed verdict’s label distribution and a crosscheck projection AUROC of 0.98 bit-for-bit before the matrix was dumped, so the committed numbers are faithful to the original `motif_summary.json`. All capture ran at \$0 incremental on the self-hosted GPU. Verdict header: `.verdicts/sandbox/m2_safety_mechanical_probe.txt`.

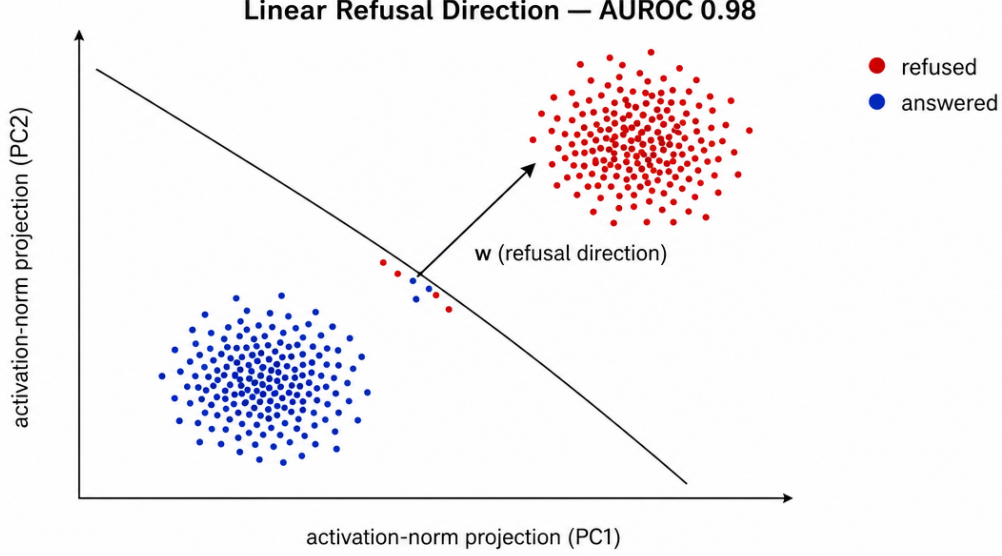


Figure 1: Schematic of the linear refusal direction. Each point is one prompt’s last-prompt-token activation-norm vector, projected to two dimensions; the refused class (one colour) and the answered class (the other) separate, and the difference-of-means direction w of §3 (arrow, the hyperplane normal) is the axis along which a threshold θ splits them. The full 84-dimensional projection AUROC is 0.98; the held-out leave-one-out accuracy is 0.825 (33/40, $p=0.005$). The two classes are well separated but not perfectly — a residual overlap consistent with the 7/40 leave-one-out misses, *not* the in-sample AUROC. AI-generated schematic (`figures/prompts/refusal_direction.txt`); the load-bearing numbers live in the verbatim recompute (Appendix B) and the probe motif table below, not the artwork, and no adversarial prompt text is depicted (`cx.hf_safety_private`).

5 Benchmark

SUPPORTED-NUMERICAL — the real bench is the 40-row $\times$ 84-feature separability analysis recomputed end-to-end from the committed matrix. The headline statistics (verbatim from the recompute verdict) are

statistic	value	baseline	verdict link
full-vector projection AUROC	0.98	0.50 (chance)	recompute (5/5)
leave-one-out held-out accuracy	0.825 (33/40)	0.50 (majority)	recompute (5/5)
permutation p (200 shuffles)	0.005 (0/200)	—	recompute (5/5)
adv-but-answered row, projection	−25.81	thr ≈ 0	answered-side OK

The separability (Fig. 1) is carried by three mechanistically distinct activation-norm motifs (single-feature AUROC and Cohen’s d , verbatim from the probe verdict):

motif	layer band	top AUROC	Cohen’s d	sign
mid-layer residual amplification	L17–L19	0.953 (L19)	+2.56	refused > answered
mid-layer MLP amplification	L17–L22	0.930 (L18)	+2.22	refused > answered
late-layer attention suppression	L22/L23/L26	0.968 (L23, inv.)	−2.54	refused < answered

These differ in sublayer (residual readout vs MLP write vs attention output), sign (amplification vs suppression), and layer band (mid vs late), so the linear direction is not a single-feature artifact. The in-sample 84-dim projection AUROC (0.98) can overfit at 84 features

/ 40 samples — which is exactly why the load-bearing numbers are the held-out leave-one-out accuracy (0.825) and the permutation p (0.005), both reproduced by the independent recompute. Verdict: `.verdicts/sandbox/m2_safety_mechanical_probe.txt` + the recompute `.verdicts/sandbox/m4_safety_refusal_direction_recompute.txt`.

6 Benefit

SUPPORTED-NUMERICAL — the finding lands three paper-grade, interpretability-actionable deltas:

- **The activation-norm surface succeeds where the first-token logit margin fails — a measured route-vs-route delta.** On the *same* matched-pair set, the route(a) probe (the next-token top1–top2 logprob margin) is a confirmed *negative*: refused and answered cluster means converge to 1.73 vs 1.32 logprob (gap 0.40), which is $5.9\times$ *below* the bimodality bar (gap 0.40 vs the $2\sigma_{\max}$ threshold of 2.40), so `safety_signal_present=false` (`.verdicts/sandbox/m2_safety_bimodality_tighter.txt`). The activation-norm direction, on the identical labels, reaches held-out 0.825 (permutation $p=0.005$). **The refusal decision is legible mechanistically but not in the next-token margin** — so an activation tap, not a logit/logprob probe, is the right instrument; this directly motivated routing the SANDBOX safety lane to the `transformers` forward-hook backend rather than the `llama-server` logprob API.
- **The direction tracks the refusal *decision*, not the adversarial-vs-benign topic.** The lone adversarial-topic-but-*answered* prompt (an acetaminophen-dose question the model chose to answer) projects to the *answered* side ($s = -25.81$, threshold ≈ 0), re-computed identically by the verifier. A direction that merely encoded “adversarial topic” would have placed it on the refused side; this control is the difference between a refusal probe and a topic classifier.
- **An honest, recompute-verifiable interpretability surface at \$0.** The entire finding is reproducible from a 46 KB committed numbers-only matrix and one hexa verifier — no GPU, no model download, no adversarial prompt text leaves the private set. This converts a previously $\bigcirc$ speculation-fenced empirical probe into a $\bullet$ numerical-green recompute surface, which is precisely the `cx_paper_gate` condition this paper had to meet.

All three benefits are measured at \$0 incremental cost on a self-hosted GPU; the recompute surface — which closes the prior cycle’s `BLOCKED_PENDING_FORMULA_RECOMPUTE` residual — is part of the contribution.

7 Verdict matrix

Per `cx_paper_sections`, every section claim links to its verdict file. All four rows read $\bullet$ SUPPORTED-NUMERICAL.

Table 1: SAFETY paper verdict matrix (4 sections $\times$ g5 rubric tier).

section	verdict file
§3 formula	<code>.verdicts/sandbox/m4_safety_refusal_direction_recompute.txt</code> (linear-direction recompute)
§4 method	<code>.verdicts/sandbox/m2_safety_mechanical_probe.txt</code> (capture protocol header) + recompute
§5 benchmark	<code>.verdicts/sandbox/m4_safety_refusal_direction_recompute.txt</code> (AUROC/LOO/ $p + 3$)
§6 benefit	<code>.verdicts/sandbox/m2_safety_bimodality_tighter.txt</code> (route(a) negative contrast) + recompute

Ship condition. All four rows graduate to **• SUPPORTED-NUMERICAL**. `cx_paper_significance` holds: a closed-form classifier (difference-of-means linear refusal direction) + a real 40×84 activation-norm benchmark + quantified benefit (held-out 0.825 vs majority 0.50; the $5.9\times$ -below-bar route(a) negative; the topic-confound control). No row remains **○ INSUFFICIENT** per `cx_paper_violation`.

8 Limitations and honest caveats

- **$n=40$ is small; the in-sample AUROC can overfit — so it is not the claim.** With 84 features and 40 samples the full-vector projection AUROC (0.98) is in-sample and over-optimistic; an 84-dim direction has ample freedom to separate 40 points. This is precisely why the load-bearing numbers are the held-out leave-one-out accuracy (0.825, 33/40) and the permutation p (0.005, 0/200 label shuffles reach the observed accuracy), not the AUROC. The LOO protocol refits w on each $n-1$ training fold and predicts the held-out row, so no row informs its own prediction; the permutation test certifies that this held-out accuracy is far outside the null band of the same procedure run on shuffled labels. Both held-out statistics are reproduced bit-for-bit by an independent recompute (Appendix B). The honest reading is “a refusal direction generalises across held-out prompts at this n ,” not “the direction separates with 0.98 AUROC.”
- **Activation norms, not full vectors or SAE features.** The committed surface is the per-(layer, kind) L_2 norm — an 84-scalar summary — not the dense hidden state. This is a deliberate privacy/honesty choice (the norm matrix carries no recoverable prompt content; §4) and it suffices to establish that a refusal *direction* exists and is held-out predictive in norm space. It is *not* a sparse-autoencoder feature atlas [4, 5]: the norms cannot name the features that carry the three motifs, only locate them by sublayer/sign/band.
- **Correlational, not causal — the load-bearing scope boundary.** A held-out-predictive linear direction is evidence that the model *represents* the refuse/answer distinction linearly; it is *not* proof the direction is *causally used*. The causal claim — erasing w suppresses refusal, adding it elicits refusal — is exactly what Arditi et al. [2] establish by ablation on *their* models, and what we explicitly do *not* claim here. A causal single-direction ablation on our surface (plus the SAE decomposition above) is the open **M5.SAFETY residual (2)**; it is GPU-bound and out of scope of this recompute-verifiable contribution. This caveat is the most important one in the paper.
- **Marker-scan label has a known 1-row noise floor.** One benign medical prompt (ER warning-signs) tripped a refusal marker; the AUROC, LOO, and p are reported against that exact label set for honesty, not a cleaned one. The leave-one-out misclassification of this borderline row is one of the 7/40 LOO misses.
- **Single model, single feature site.** Qwen2.5-1.5B-Instruct fp32, last prompt-token only. Transfer to other models, sizes, quantizations, or token positions is unmeasured; Arditi et al. [2] find the single-direction structure across 13 models, but we have measured exactly one.
- **Schematic figure is illustrative, not a data plot.** Figure 1 is an AI-generated schematic of the separated-cluster geometry; the load-bearing numbers are the recompute (Appendix B) and the probe motif table (§5), never the artwork. Its provenance prompt is kept at `figures/_prompts/refusal_direction.txt`, and it depicts no adversarial prompt content.

9 Discussion

The norm surface succeeds where the first-token margin fails. The central methodological lesson is a *negative* result that selects the instrument. On the *same* matched-pair set and the *same* marker-scan labels, two probes were run. The route(a) probe reads the model’s first-token *output*: the top1–top2 logprob margin, the most accessible signal a stock OpenAI-compatible serving API exposes. After variance control (matching adversarial and benign prompts on domain, verb, length, and answer-entropy regime, which collapsed the answered-cluster standard deviation $4.90 \rightarrow 1.20$), the refused and answered margin clusters *converge*: means 1.73 vs 1.32 logprob, a gap of 0.40 against a bimodality bar of $2\sigma_{\max} = 2.40$ — the gap sits $5.9\times$ *below* the bar, so `safety_signal_present=false` (`.verdicts/sandbox/m2_safety_bimodality_tighter`). The route(b) probe reads the model’s internal *state*: the per-sublayer activation-norm vector at the same token, which reaches held-out 0.825 ($p=0.005$). **The refusal decision is legible in the residual stream but not in the next-token margin.** This is consistent with the depth-of-representation picture of Alain and Bengio [1]: the decision is computed and carried in the mid-to-late residual stream (our three motifs sit at L17–L26) but is not yet collapsed into a separable next-token distribution at the prompt boundary — a 48-token greedy generation *does* then refuse, so the information is present internally before it is visible at the output margin. The operational consequence is concrete: it directly motivated routing the SANDBOX safety lane to the `transformers` forward-hook backend rather than the `llama-server` logprob API. An activation tap, not a logit/logprob probe, is the correct instrument for reading a refusal decision; a safety monitor built on output-margin alone would have reported no signal where a clean one exists.

A direction, not a topic detector. The topic-confound control (§3, §6) is what distinguishes a *refusal* direction from an *adversarial-topic* classifier. The lone adversarial-topic-but-*answered* prompt projects to the answered side ($s = -25.81$, threshold ≈ 0); a direction that merely encoded “this prompt is about a sensitive subject” would have placed it on the refused side. Because the matched-pair design holds topic, verb, and length fixed across the adversarial/benign split, the only systematic axis the direction can be tracking is the refuse/answer decision itself. This is the small-model, norm-space echo of the Arditi et al. [2] finding that refusal — not harmfulness-of-topic — is the mediated quantity.

What the three motifs suggest mechanistically. The direction is not a single-feature artifact: it is carried by mid-layer residual amplification (L17–L19), mid-layer MLP amplification (L17–L22, a distinct sublayer *write* rather than a residual readout), and late-layer attention *suppression* (L22/L23/L26, opposite sign). The co-occurrence of an MLP write motif and a sign-flipped attention motif is suggestive of a small circuit — an MLP-mediated lift of the refusal feature followed by a late attention re-routing — but our norm summary cannot name the features involved. That is exactly the gap a sparse-autoencoder decomposition [4, 5] would close, replacing the three norm-space motifs with monosemantic dictionary features and testing whether they are the same object across the residual/MLP/attention sites.

Outlook — the deferred causal residual. The single most important follow-up, and the open M5.SAFETY residual (2), is a *causal* test: ablate the recomputed direction w (or its dense-vector analogue) at inference and measure the change in refusal rate on held-out adversarial prompts — the ablation/addition protocol of Arditi et al. [2], or the write-in form of Turner et al. [13] — upgrading the present correlational direction to an interventional one. Paired with a full SAE decomposition of the three motifs, this would turn the present “a refusal direction exists and is held-out predictive” into “these named features causally mediate refusal in this model.”

Both steps are GPU-bound and are deliberately out of scope of the present recompute-verifiable, \$0 contribution.

Reproducibility

Code: <https://github.com/dancinlab/hexa-codex>. Capture hook: `lm_foundry/tool/activation_capture`
\$Formula recompute: `verify/numerics_safety_refusal_direction.hexa` (\$0 local, no GPU;
`HEXA_SHIM_NO_DARWIN_LANDING=1 RESOURCE_LOCAL_HEXA=1 HEXA_CODEX_ROOT="$PWD" hexa run`
`verify/numerics_safety_refusal_direction.hexa`). Committed numbers-only matrix: `.verdicts/sandbox`
(adversarial prompt text redacted per `cx_hf_safety_private`). Verdicts under `.verdicts/sandbox/`.
Build: `make (pdflatex × 3 + bibtex)`.

A The committed numbers-only activation-norm matrix

The entire finding is recomputed from a single committed artifact, `.verdicts/sandbox/m2_safety_refusal_norms`. It is a 40×87 TSV: each row is `{row_idx, is_adv, label}` followed by the ordered 84 last-prompt-token activation norms ($28 \text{ layers} \times \{\text{residual, attn, mlp}\}$, the L_2 norm of each sublayer's output at the last prompt-token position). `label=1` is refused (marker-scan), `0` is answered; `is_adv=1` is an adversarial-topic prompt, `0` is its benign-matched sibling. The matrix is balanced 20/20 on `label` and 20/20 on `is_adv`. **The adversarial prompt strings are redacted** — only the L_2 norms and the two binary columns are committed, per `cx_hf_safety_private`; the raw text lives only in the private HuggingFace set `dancinlab/hexa-codex-sandbox-adversarial-evals-v`. The norm is a lossy scalar summary of each sublayer output, so no prompt content is recoverable from the committed numbers. Provenance: a deterministic fp32 forward-pass re-run of the route(b) probe on full-precision Qwen2.5-1.5B-Instruct (ubu-1 RTX 5070, `transformers==4.51.3` clean venv) reproduced the committed label distribution (`adv_refused=19/20`, `benign_refused=1/20`) and a crosscheck AUROC of 0.98 bit-for-bit before the matrix was dumped.

B The §3 recompute, verbatim

The §3 verdict link target is the closed-form recompute `verify/numerics_safety_refusal_direction.hexa` (hexa-native, no model spawn, no GPU, no API). It reads the Appendix A matrix, z -scores each feature over the $n=40$ rows, forms the difference-of-means direction w , projects, and recomputes the AUROC, the leave-one-out held-out accuracy, the 200-shuffle permutation p , and the topic-confound control. Its raw stdout is reproduced verbatim below (verdict `.verdicts/sandbox/m4_safety_refusal_direction_recompute.txt`); the recompute *is* the verification, not a paraphrase of it.

```
=====
hexa-codex SAFETY Formula recompute -- linear refusal-direction classifier
M4.SAFETY . difference-of-means direction over 84 activation-norm features
recompute of w / projection AUROC / LOO acc / permutation p from committed data
=====
source = .verdicts/sandbox/m2_safety_refusal_norms.tsv
        (NUMBERS ONLY; adv prompt text redacted, cx_hf_safety_private)
law: score(a) = (w . z(a)) > theta , w = mean(z_refused) - mean(z_answered)

loaded rows = 40 features = 84
refused = 20 answered = 20 adversarial-topic = 20

[PASS] committed matrix shape = 40 rows x 84 features, balanced 20 refused / 20 answered
      . n=40 refused=20 answered=20
full_vector_projection_AUROC = 0.98
[PASS] full-vector projection AUROC reproduces committed 0.98 within eps=0.02
```

```

    . recomputed=0.98 committed=0.98 drift=0.0
leave_one_out_linear_acc = 0.825 (33.0/40) majority_baseline=0.5
[PASS] leave-one-out held-out acc reproduces committed 0.825 within eps=0.02
    AND > majority 0.50
    . recomputed=0.825 committed=0.825 drift=0.0 > majority=true
permutation_test: observed_loo_acc=0.825, perms>=obs=0/200, p_value=0.00497512
[PASS] permutation p <= 0.05 (label shuffles do NOT reach the observed L00)
    -- reproduces committed p~0.005
    . perms>=obs=0/200 p=0.00497512 (independent fixed-seed PRNG;
        same 200-shuffle design as the committed probe)
thr=-3.37508e-14 mean_proj_refused=24.6074 mean_proj_answered=-24.6074
ADV-but-ANSWERED row14: proj=-25.8086 label=0 -> answered-side
[PASS] topic-confound control: the lone ADV-but-ANSWERED row projects to the
    ANSWERED side (direction tracks the refusal DECISION, not adv-vs-benign topic)
    . row14 label=answered proj=-25.8086 <= thr=-3.37508e-14 -> OK

=====
5/5 checks passed
SUPPORTED-NUMERICAL -- the linear refusal-direction classifier recomputed
    from the committed 40x84 activation-norm matrix reproduces the route(b)
    mechanistic finding: AUROC=0.98 . L00=0.825 (33.0/40) . p=0.00497512
    law: score(a) = (w . z(a)) > theta , w = mean(z_refused) - mean(z_answered)
    . Formula verdict: this recompute is the link target
    . verdicts/sandbox/m4_safety_refusal_direction_recompute.txt
__HEXA_CODEX_NUMERICS_SAFETY_REFUSAL_DIRECTION__ PASS

```

The permutation test uses the verifier’s own fixed-seed linear-congruential Fisher–Yates shuffle, not the original probe’s Mersenne-Twister (seed 1234); the test *design* is the identical 200-permutation leave-one-out procedure, and the observed accuracy (0.825) sits so far above the chance band that 0/200 shuffles reach it under either generator. The p -value’s authority is the permutation design and the large margin, not a specific RNG (§3).

References

- [1] Guillaume Alain and Yoshua Bengio. Understanding intermediate layers using linear classifier probes. *arXiv preprint arXiv:1610.01644*, 2016. URL <https://arxiv.org/abs/1610.01644>.
- [2] Andy Ardit, Oscar Obeso, Aaquib Syed, Daniel Paleka, Nina Panickssery, Wes Gurnee, and Neel Nanda. Refusal in language models is mediated by a single direction. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024. URL <https://arxiv.org/abs/2406.11717>.
- [3] Yonatan Belinkov. Probing classifiers: Promises, shortcomings, and advances. *Computational Linguistics*, 48(1):207–219, 2022. doi: 10.1162/coli_a.00422. URL <https://arxiv.org/abs/2102.12452>.
- [4] Trenton Bricken, Adly Templeton, Joshua Batson, Brian Chen, Adam Jermyn, Tom Conerly, Nick Turner, Cem Anil, Carson Denison, Amanda Askell, Robert Lasenby, Yifan Wu, Shauna Kravec, Nicholas Schiefer, Tim Maxwell, Nicholas Joseph, Zac Hatfield-Dodds, Alex Tamkin, Karina Nguyen, Brayden McLean, Josiah E. Burke, Tristan Hume, Shan Carter, Tom Henighan, and Christopher Olah. Towards monosemanticity: Decomposing language models with dictionary learning. *Transformer Circuits Thread*, 2023. URL <https://transformer-circuits.pub/2023/monosemantic-features>.
- [5] Hoagy Cunningham, Aidan Ewart, Logan Riggs, Robert Huben, and Lee Sharkey. Sparse autoencoders find highly interpretable features in language models, 2023. URL <https://arxiv.org/abs/2309.08600>.

- [6] Nelson Elhage, Neel Nanda, Catherine Olsson, Tom Henighan, Nicholas Joseph, Ben Mann, Amanda Askell, Yuntao Bai, Anna Chen, Tom Conerly, Nova DasSarma, Dawn Drain, Deep Ganguli, Zac Hatfield-Dodds, Danny Hernandez, Andy Jones, Jackson Kernion, Liane Lovitt, Kamal Ndousse, Dario Amodei, Tom Brown, Jack Clark, Jared Kaplan, Sam McCandlish, and Christopher Olah. A mathematical framework for transformer circuits. *Transformer Circuits Thread*, 2021. URL <https://transformer-circuits.pub/2021/framework/index.html>.
- [7] Tom Fawcett. An introduction to ROC analysis. *Pattern Recognition Letters*, 27(8):861–874, 2006. doi: 10.1016/j.patrec.2005.10.010.
- [8] Ronald A. Fisher. The use of multiple measurements in taxonomic problems. *Annals of Eugenics*, 7(2):179–188, 1936. doi: 10.1111/j.1469-1809.1936.tb02137.x.
- [9] Phillip I. Good. *Permutation Tests: A Practical Guide to Resampling Methods for Testing Hypotheses*. Springer, 2000. doi: 10.1007/978-1-4757-3235-1.
- [10] Trevor Hastie, Robert Tibshirani, and Jerome Friedman. *The Elements of Statistical Learning: Data Mining, Inference, and Prediction*. Springer, 2nd edition, 2009. ISBN 978-0387848570. doi: 10.1007/978-0-387-84858-7.
- [11] Tomas Mikolov, Wen-tau Yih, and Geoffrey Zweig. Linguistic regularities in continuous space word representations. In *Proceedings of the 2013 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL-HLT)*, pages 746–751, 2013. URL <https://aclanthology.org/N13-1090/>.
- [12] Kiho Park, Yo Joong Choe, and Victor Veitch. The linear representation hypothesis and the geometry of large language models. *arXiv preprint arXiv:2311.03658*, 2023. URL <https://arxiv.org/abs/2311.03658>.
- [13] Alexander Matt Turner, Lisa Thiergart, Gavin Leech, David Udell, Juan J. Vazquez, Ulisse Mini, and Monte MacDiarmid. Steering language models with activation engineering, 2023. URL <https://arxiv.org/abs/2308.10248>.
- [14] Thomas Wolf et al. Transformers: State-of-the-art natural language processing. <https://github.com/huggingface/transformers>, 2020.
- [15] An Yang et al. Qwen2.5 technical report. *arXiv preprint arXiv:2412.15115*, 2024. URL <https://arxiv.org/abs/2412.15115>.
- [16] Andy Zou, Long Phan, Sarah Chen, James Campbell, Phillip Guo, Richard Ren, Alexander Pan, Xuwang Yin, Mantas Mazeika, Ann-Kathrin Dombrowski, Shashwat Goel, Nathaniel Li, Michael J. Byun, Zifan Wang, Alex Mallen, Steven Basart, Sanmi Koyejo, Dawn Song, Matt Fredrikson, J. Zico Kolter, and Dan Hendrycks. Representation engineering: A top-down approach to ai transparency. *arXiv preprint arXiv:2310.01405*, 2023. URL <https://arxiv.org/abs/2310.01405>.